

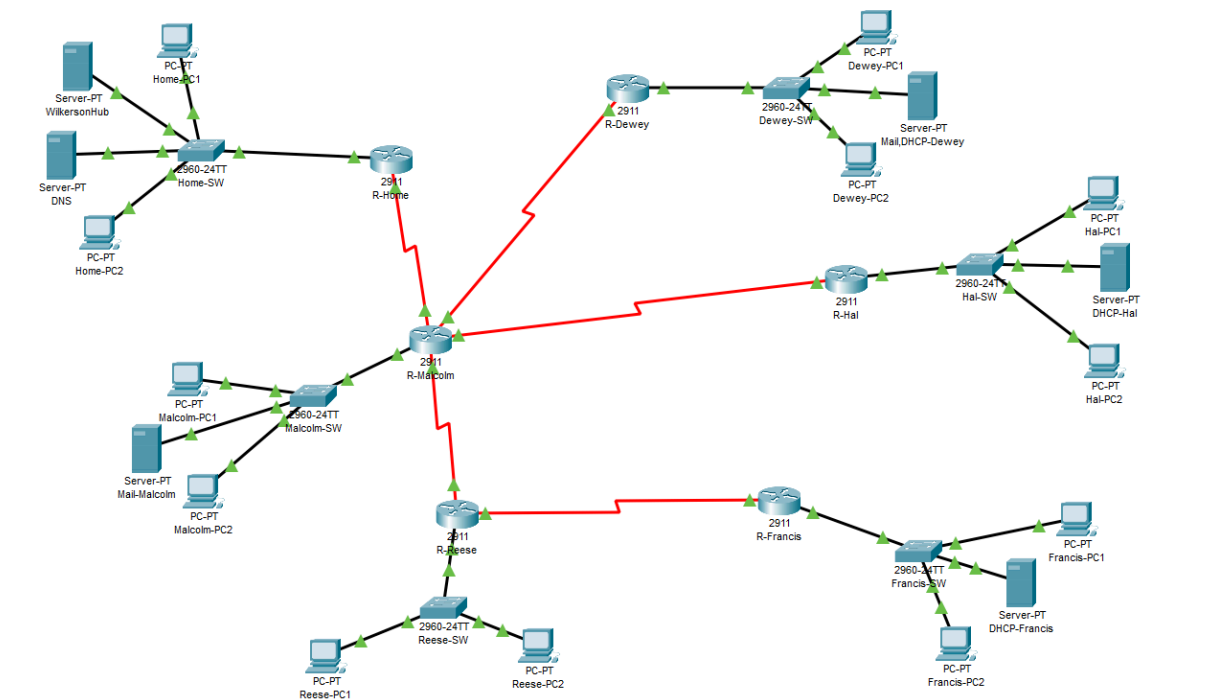
Network Infrastructure Design & Implementation

CSE421 - Computer Networks

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Deliverable 1: Network Topology Diagram

The topology diagram below shows the complete network implemented in Cisco Packet Tracer, including all routers, switches, servers, and end devices with their labels.



Devices used: 6 × Cisco 2911 Routers, 6 × Cisco 2960 Switches, 6 × Server-PT, 12 × PC-PT. WAN links use Serial DCE/DTE cables. LAN links use Copper Straight-Through cables.

Deliverable 2: Router Configuration Commands

R-Home

```
enable
configure terminal
hostname R-Home

interface GigabitEthernet0/0
ip address 13.14.9.129 255.255.255.224
no shutdown

interface Serial0/0/0
ip address 13.14.9.161 255.255.255.252
clock rate 64000
no shutdown

ip route 0.0.0.0 0.0.0.0 13.14.9.162
ip route 13.14.6.0 255.255.254.0 13.14.9.162
ip route 13.14.4.0 255.255.254.0 13.14.9.162 25

end
write memory
```

R-Malcolm (Hybrid Router - RIP + Static)

```
enable
configure terminal
hostname R-Malcolm

interface GigabitEthernet0/0
ip address 13.14.0.1 255.255.252.0
no shutdown

interface Serial0/0/0
ip address 13.14.9.162 255.255.255.252
no shutdown

interface Serial0/0/1
ip address 13.14.9.165 255.255.255.252
clock rate 64000
no shutdown

interface Serial0/1/0
ip address 13.14.9.169 255.255.255.252
clock rate 64000
no shutdown

interface Serial0/1/1
```

```
ip address 13.14.9.173 255.255.255.252
clock rate 64000
no shutdown

ip dhcp excluded-address 13.14.0.1 13.14.0.10
ip dhcp pool Malcolm-Pool
network 13.14.0.0 255.255.252.0
default-router 13.14.0.1
dns-server 13.14.9.131

ip route 13.14.9.128 255.255.255.224 13.14.9.161
ip route 13.14.6.0 255.255.254.0 13.14.9.174
ip route 13.14.4.0 255.255.254.0 13.14.9.170

router rip
version 2
no auto-summary
network 13.0.0.0
passive-interface GigabitEthernet0/0
passive-interface Serial0/0/0
passive-interface Serial0/1/0

end
write memory
```

R-Hal

```
enable
configure terminal
hostname R-Hal

interface GigabitEthernet0/0
ip address 13.14.9.1 255.255.255.128
no shutdown

interface Serial0/0/0
ip address 13.14.9.166 255.255.255.252
no shutdown

ip dhcp excluded-address 13.14.9.1 13.14.9.10
ip dhcp pool Hal-Pool
network 13.14.9.0 255.255.255.128
default-router 13.14.9.1
dns-server 13.14.9.131

ip helper-address 13.14.9.2

ip route 13.14.9.128 255.255.255.224 13.14.9.165
ip route 13.14.6.0 255.255.254.0 13.14.9.165
ip route 13.14.4.0 255.255.254.0 13.14.9.165
```

```
router rip
version 2
no auto-summary
network 13.0.0.0
passive-interface GigabitEthernet0/0

end
write memory
```

R-Reese

```
enable
configure terminal
hostname R-Reese

interface GigabitEthernet0/0
ip address 13.14.8.1 255.255.255.0
no shutdown

interface Serial0/0/0
ip address 13.14.9.174 255.255.255.252
no shutdown

interface Serial0/0/1
ip address 13.14.9.177 255.255.255.252
clock rate 64000
no shutdown

ip dhcp excluded-address 13.14.8.1 13.14.8.10
ip dhcp pool Reese-Pool
network 13.14.8.0 255.255.255.0
default-router 13.14.8.1
dns-server 13.14.9.131

ip route 13.14.9.128 255.255.255.224 13.14.9.173
ip route 13.14.0.0 255.255.252.0 13.14.9.173
ip route 13.14.9.0 255.255.255.128 13.14.9.173
ip route 13.14.6.0 255.255.254.0 13.14.9.178
ip route 13.14.4.0 255.255.254.0 13.14.9.173

router rip
version 2
no auto-summary
network 13.0.0.0
passive-interface GigabitEthernet0/0
passive-interface Serial0/0/1

end
write memory
```

R-Francis

```
enable
configure terminal
hostname R-Francis

interface GigabitEthernet0/0
ip address 13.14.6.1 255.255.254.0
no shutdown

interface Serial0/0/0
ip address 13.14.9.178 255.255.255.252
no shutdown

ip helper-address 13.14.6.2

ip route 0.0.0.0 0.0.0.0 13.14.9.177
ip route 13.14.9.128 255.255.255.224 13.14.9.177
ip route 13.14.0.0 255.255.252.0 13.14.9.177
ip route 13.14.4.0 255.255.254.0 13.14.9.177
ip route 13.14.8.0 255.255.255.0 13.14.9.177
ip route 13.14.9.0 255.255.255.128 13.14.9.177

end
write memory
```

R-Dewey

```
enable
configure terminal
hostname R-Dewey

interface GigabitEthernet0/0
ip address 13.14.4.1 255.255.254.0
no shutdown

interface Serial0/0/0
ip address 13.14.9.170 255.255.255.252
no shutdown

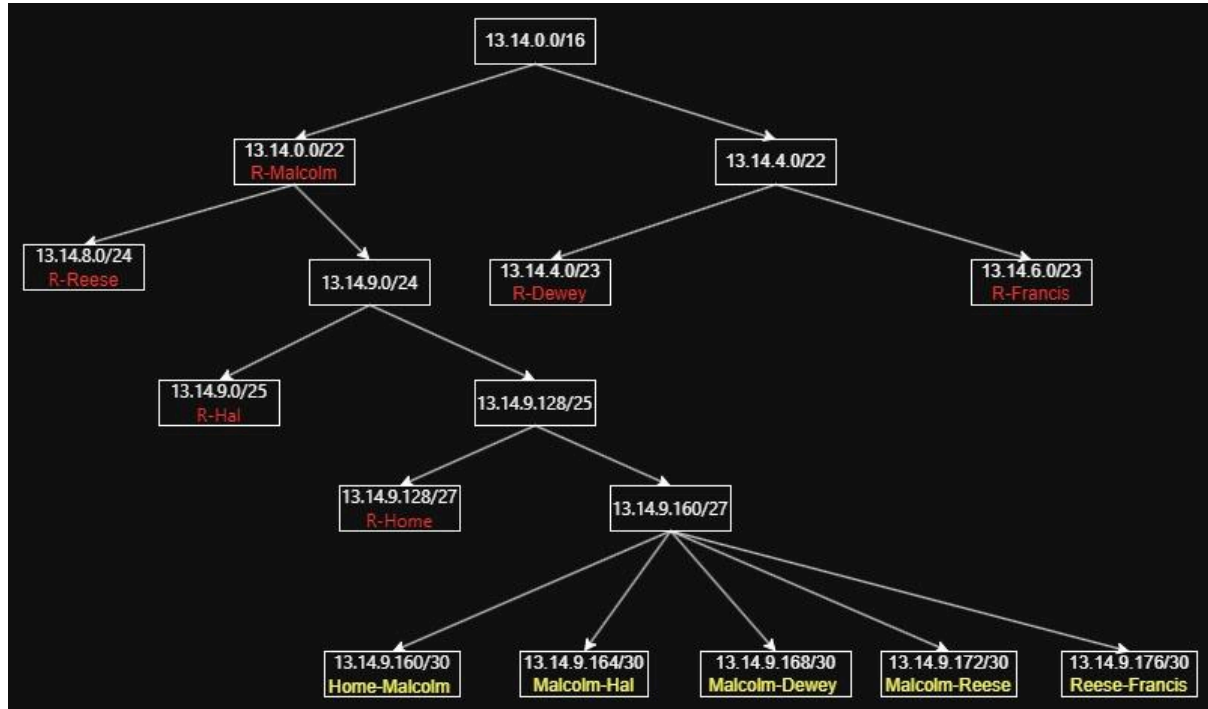
ip helper-address 13.14.4.2

ip route 0.0.0.0 0.0.0.0 13.14.9.169

end
write memory
```

Deliverable 3: Subnetting Plan

VLSM Tree



Subnet Allocation Table

Network Zone	Subnet	Mask	Hosts Needed	Usable Hosts
Malcolm's School Lab	13.14.0.0/22	255.255.252.0	840	1022
Dewey's Creative Lab	13.14.4.0/23	255.255.254.0	300	510
Francis's Military Acad.	13.14.6.0/23	255.255.254.0	390	510
Reese's Kitchen Zone	13.14.8.0/24	255.255.255.0	210	254
Hal's Workplace	13.14.9.0/25	255.255.255.128	72	126
Family Home Network	13.14.9.128/27	255.255.255.224	25	30
WAN: Home-Malcolm	13.14.9.160/30	255.255.255.252	2	2
WAN: Malcolm-Hal	13.14.9.164/30	255.255.255.252	2	2
WAN: Malcolm-Dewey	13.14.9.168/30	255.255.255.252	2	2
WAN: Malcolm-Reese	13.14.9.172/30	255.255.255.252	2	2
WAN: Reese-Francis	13.14.9.176/30	255.255.255.252	2	2

Deliverable 4: IP Address Table

Router LAN Interfaces and Default Gateways

Router	Interface	IP Address	Subnet Mask	Default Gateway
R-Home	GigabitEthernet0/0	13.14.9.129	255.255.255.224	13.14.9.129
R-Malcolm	GigabitEthernet0/0	13.14.0.1	255.255.252.0	13.14.0.1
R-Hal	GigabitEthernet0/0	13.14.9.1	255.255.255.128	13.14.9.1
R-Reese	GigabitEthernet0/0	13.14.8.1	255.255.255.0	13.14.8.1
R-Francis	GigabitEthernet0/0	13.14.6.1	255.255.254.0	13.14.6.1
R-Dewey	GigabitEthernet0/0	13.14.4.1	255.255.254.0	13.14.4.1

Router WAN (Serial) Interfaces

Router	Interface	IP Address	Subnet Mask	Role
R-Home	Serial0/0/0	13.14.9.161	255.255.255.252	DCE
R-Malcolm	Serial0/0/0	13.14.9.162	255.255.255.252	DTE
R-Malcolm	Serial0/0/1	13.14.9.165	255.255.255.252	DCE
R-Hal	Serial0/0/0	13.14.9.166	255.255.255.252	DTE
R-Malcolm	Serial0/1/0	13.14.9.169	255.255.255.252	DCE
R-Dewey	Serial0/0/0	13.14.9.170	255.255.255.252	DTE
R-Malcolm	Serial0/1/1	13.14.9.173	255.255.255.252	DCE
R-Reese	Serial0/0/0	13.14.9.174	255.255.255.252	DTE
R-Reese	Serial0/0/1	13.14.9.177	255.255.255.252	DCE
R-Francis	Serial0/0/0	13.14.9.178	255.255.255.252	DTE

Server IP Addresses

Server Name	Network	IP Address	Subnet Mask	Gateway
WilkersonHub (Web)	Home	13.14.9.130	255.255.255.224	13.14.9.129
DNS-Server	Home	13.14.9.131	255.255.255.224	13.14.9.129
MailMalcolm (Email)	Malcolm	13.14.0.2	255.255.252.0	13.14.0.1
DHCP-Francis	Francis	13.14.6.2	255.255.254.0	13.14.6.1
DHCP-Hal	Hal	13.14.9.2	255.255.255.128	13.14.9.1

Server Name	Network	IP Address	Subnet Mask	Gateway
MailDewey (Email+DHCP)	Dewey	13.14.4.2	255.255.254.0	13.14.4.1

End Device IP Addresses

Device	Network	IP Address	Subnet Mask	Gateway	Method
Home-PC1	Home	13.14.9.132	255.255.255.224	13.14.9.129	Static
Home-PC2	Home	13.14.9.133	255.255.255.224	13.14.9.129	Static
Malcolm-PC1	Malcolm	13.14.0.11	255.255.252.0	13.14.0.1	DHCP
Malcolm-PC2	Malcolm	13.14.0.12	255.255.252.0	13.14.0.1	DHCP
Hal-PC1	Hal	13.14.9.11	255.255.255.128	13.14.9.1	DHCP
Hal-PC2	Hal	13.14.9.12	255.255.255.128	13.14.9.1	DHCP
Reese-PC1	Reese	13.14.8.11	255.255.255.0	13.14.8.1	DHCP
Reese-PC2	Reese	13.14.8.12	255.255.255.0	13.14.8.1	DHCP
Francis-PC1	Francis	13.14.6.11	255.255.254.0	13.14.6.1	DHCP
Francis-PC2	Francis	13.14.6.12	255.255.254.0	13.14.6.1	DHCP
Dewey-PC1	Dewey	13.14.4.11	255.255.254.0	13.14.4.1	DHCP
Dewey-PC2	Dewey	13.14.4.12	255.255.254.0	13.14.4.1	DHCP

Deliverable 5: Base Network Address Derivation

The base network address was derived from the Student ID of the last group member, Sadat Hossain (22201314), following the method specified in the assignment.

Step	Value
Student ID of last member	22201314
Extract last 4 digits	1314
Split into two 2-digit pairs	13 14
First octet	13
Second octet	14
Third and fourth octets	0.0
Base Network Address	13.14.0.0 / 16

Deliverable 6: Routing and DHCP Summary

DHCP Configuration

Type	Network	Configured On
Router-Based DHCP	Malcolm's School Lab	R-Malcolm (DHCP pool on router)
Router-Based DHCP	Reese's Kitchen Zone	R-Reese (DHCP pool on router)
Server-Based DHCP	Francis's Military Academy	DHCP-Francis server + ip helper-address on R-Francis
Server-Based DHCP	Hal's Workplace	DHCP-Hal server + ip helper-address on R-Hal
Server-Based DHCP	Dewey's Creative Lab	MailDewey server + ip helper-address on R-Dewey

Static Routing

Router	Static Routes Configured
R-Home	Default static route (0.0.0.0/0) toward Malcolm; standard static route to Francis (13.14.6.0/23); floating static route to Dewey (13.14.4.0/23) with AD=25
R-Malcolm	Static route to Home (13.14.9.128/27); static route to Francis (13.14.6.0/23); static route to Dewey (13.14.4.0/23)
R-Reese	Static routes to Francis, Home, Dewey, and Hal - all via R-Malcolm or R-Francis
R-Francis	Static routes to all other networks via R-Reese
R-Dewey	Default static route (0.0.0.0/0) toward R-Malcolm
R-Hal	Static routes to Home, Francis, and Dewey networks via R-Malcolm

Dynamic Routing (RIP v2)

Router	Protocol	Active RIP Interfaces
R-Malcolm	RIP version 2	Serial0/0/1 (to Hal), Serial0/1/1 (to Reese)
R-Hal	RIP version 2	Serial0/0/0 (to Malcolm)
R-Reese	RIP version 2	Serial0/0/0 (to Malcolm)

Hybrid Router (R-Malcolm)

R-Malcolm is configured as the hybrid router. It runs RIP version 2 dynamically with R-Hal (via Serial0/0/1) and R-Reese (via Serial0/1/1), while simultaneously maintaining static routes to the Family Home Network, Francis's Military Academy, and Dewey's Creative Lab. The interfaces connecting to Home and Dewey are set as passive in RIP to prevent unnecessary RIP advertisements on those static-only links.

Special Routing Notes

Route Type	Detail
Default Static Route	Configured on R-Home. Forwards all unknown traffic toward R-Malcolm (next-hop 13.14.9.162).
Standard Static Routes	Configured between R-Home and Francis's Military Academy network (bidirectional reachability).
Floating Static Route	Configured on R-Home toward Dewey's Lab (13.14.4.0/23) via R-Malcolm with Administrative Distance = 25. This route only becomes active if the primary path fails.
Direct Router Links	Francis <-> Reese: R-Francis Se0/0/0 <-> R-Reese Se0/0/1. Hal <-> Malcolm: R-Hal Se0/0/0 <-> R-Malcolm Se0/0/1.